

Project Report

Sentiment Analysis on Movie Reviews

Task 1 - DLBAIPNLP01 Project: NLP

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Dataset: Stanford Large Movie Review Dataset (IMDB)

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List of Abbreviations

Abbreviation	Meaning
BERT	Bidirectional Encoder Representations from Transformers
DL	Deep Learning
F1	Harmonic mean of precision and recall
GPU	Graphics Processing Unit
IMDB	Internet Movie Database
NLP	Natural Language Processing
SST-2	Stanford Sentiment Treebank, binary version
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency

0.Abstract

This paper discusses the creation and assessment of a Natural Language Processing (NLP) solution for the binary sentiment analysis of movie critiques. The solution was capable of distinguishing reviews as either positive or negative by traditional machine learning methods as well as state-of-the-art transformerbased deep learning architectures. The primary data source was the publicly available Stanford Large Movie Review Dataset (IMDB dataset) which contains 50, 000 movie reviews with labels and shares of training and test sets are equal.

Fully Functional NLP pipeline designed, produced, managed and operated text preprocessing, extraction of TF-IDF features, supervised model learning, and comparative performance assessment are the main components. Naive Bayes, Support Vector Machines (SVM), and other traditional machine learning methodologies were trained there were small, medium, and entire training samples so that the model performance could be gauged as a function of the data set size.

Besides, a feed-forward neural network was implemented using TF-IDF features as a deep learning baseline.

As for further analysis, a pretrained DistilBERT transformer model fine-tuned on the SST2 benchmark dataset was run on the entire IMDB test set of 25, 000 reviews. Accuracy, precision, recall, and F1score were the metrics used by the evaluation. Results indicate that transformer-based contextual language models drastically surpass classical TF-IDF-based techniques and that DistilBERT exemplified the best overall results.

Moreover, the project includes graphical presentation, confusion matrices, runtime comparisons, and practical testing of novel reviews. According to the research, the balance between computational efficiency and predictive performance must be considered, and while SVM is a very good lightweight baseline, DistilBERT produces much better results with sentiment classification.

What is more, the project shows a full end-to-end sentiment analysis solution and features a thorough evaluation of classical machine learning algorithms and cutting edge transformer-based NLP methods.

1. Introduction

Movie reviews provide a well-known channel for the public to express opinions about films. For film studios, streaming services, and rating compilation websites, being able to determine if a review conveys a positive or a negative sentiment is instrumental in helping market research, customer tracking, and content evaluation. The assigned project task involves creating a natural language processing (NLP) system for binary sentiment analysis of movie reviews using publicly available labelled data.

It aims at developing, implementing and assessing a supervised NLP pipeline that differentiates movie reviews into positive or negative. It includes the gathering of data, its processing, representing text in a numerical format, construction of machine learning models, evaluating the models with a completely separate dataset, and the use of the model on the new set after training is complete. The project needs continuous evaluation starting from a small dataset to the bigger ones; this condition is satisfied by the evaluation of traditional models on training samples of 1, 000 set , 10, 000 set , and 25, 000 set.

The project stages two directions of both simple and sophisticated. Traditional models are baselined by a Naive Bayes and a linear Support Vector Machine (SVM) trained on TF-IDF features. An elementary feed-forward neural network is used for training on TF-IDF features as a deep-learning alternative. A pre-trained DistilBERT sentiment-analysis pipeline is tested as the state-of-the-art transformer-based model. In the end, models are compared based on accuracy, precision, recall and F1-score obtained on the entire test dataset.

The report is structured following the outline proposed for an IU project report: objective and planning, implementation, evaluation, reflection and conclusion. The focus is mainly on the documentation of a functioning NLP system and critical assessment of the suitability of the applied methods.

2. Project Planning and Resources

Originally, a practical supervised text-classification system was what the project was about. The main development phases were getting data, preprocessing, feature coding, step-by-step model training, sophisticated model testing, final comparison and use on new example reviews. Google Colab with GPU acceleration was chosen for the transformer-based model since it is very inefficient to run DistilBERT on 25, 000 test reviews on a CPU-only laptop.

Table 1: Resources and tools used in the project

Resource / Tool	Use in the project
Python	Main programming language for the NLP pipeline
Google Colab GPU	Efficient DistilBERT inference on the full test set
Hugging Face datasets	Loading the IMDB train and test data
scikit-learn	TF-IDF vectorization, Naive Bayes, SVM and evaluation metrics
TensorFlow / Keras	Implementation of the feed-forward neural network
Transformers / PyTorch	DistilBERT sentiment-analysis pipeline
Matplotlib / seaborn	Visualizations and confusion matrices
WordCloud / NLTK	Exploratory text analysis and preprocessing

Selected tools are a great choice as they fulfill the main two project requisites very well, that is, performing few classical machine learning algorithms on the TF-IDF features for a quick estimation and making use of the transformers for very accurate sentiment classification on the whole test dataset.

The work was planned into well defined phases. The first step was to load the dataset and have a look at it. The second step was to create a preprocessing pipeline and call it a day by using both the training and testing data for testing it. Third, by training classical models one by one, the impact of the size of the training set was being observed. Fourth, for advanced level, the DistilBERT model was run on the whole test set which was possible with GPU(s). Finally, the results obtained were compared and related to the original task in consideration.

Table 2: Project phases and outputs

Project phase	Main activity	Output
Preparation	Select task, dataset and tools	Project scope and technical setup
Data handling	Load IMDB train/test split and inspect labels	Balanced labelled dataset
Preprocessing	Clean text, preserve negations and lemmatize tokens	Prepared reviews for TF-IDF models
Model training	Train Naive Bayes, SVM and neural network	Baseline and comparison models
Advanced evaluation	Run DistilBERT on the full test set with GPU	Transformer baseline metrics
Reporting	Compare metrics, reflect on limitations and document product	Final project report

3. Data Collection and Initial Conditions

The project is based on the use of the Stanford Large Movie Review Dataset, which is popularly known as the IMDB movie review dataset. It comprises 50, 000 labelled movie reviews, split equally between 25, 000 training reviews and 25, 000 test reviews. The binary labels are positive and negative. Because the class distribution is balanced, the dataset is suitable for evaluation based on accuracy; nevertheless, precision, recall, and F1-score are reported as well in order to give a more comprehensive picture of model performance.

The dataset was directly loaded via the Hugging Face datasets library. This way, there is no need to scrap the web manually and one can be sure of a reproducible source. The train/test split that came with the dataset was kept untouched, hence no test data was employed for training the model. This separation is what enables one to make a fair estimate of the generalization performance.

In order to progressively assess the model, the training set was split into three scenarios: a small sample of 1, 000 reviews, a medium sample of 10, 000 reviews and the full training set of 25, 000 reviews. The complete test set of 25, 000 reviews was utilized for evaluation in all final comparisons.

We first checked the dataset class distribution before preprocessing and model training. The IMDB dataset is balanced since it has the same number of positive and negative reviews, which means the task is a balanced binary classification problem. Having this balance means that the metrics used for evaluation such as accuracy, precision, recall, and F1-score are trustworthy and not skewed to one sentiment class.

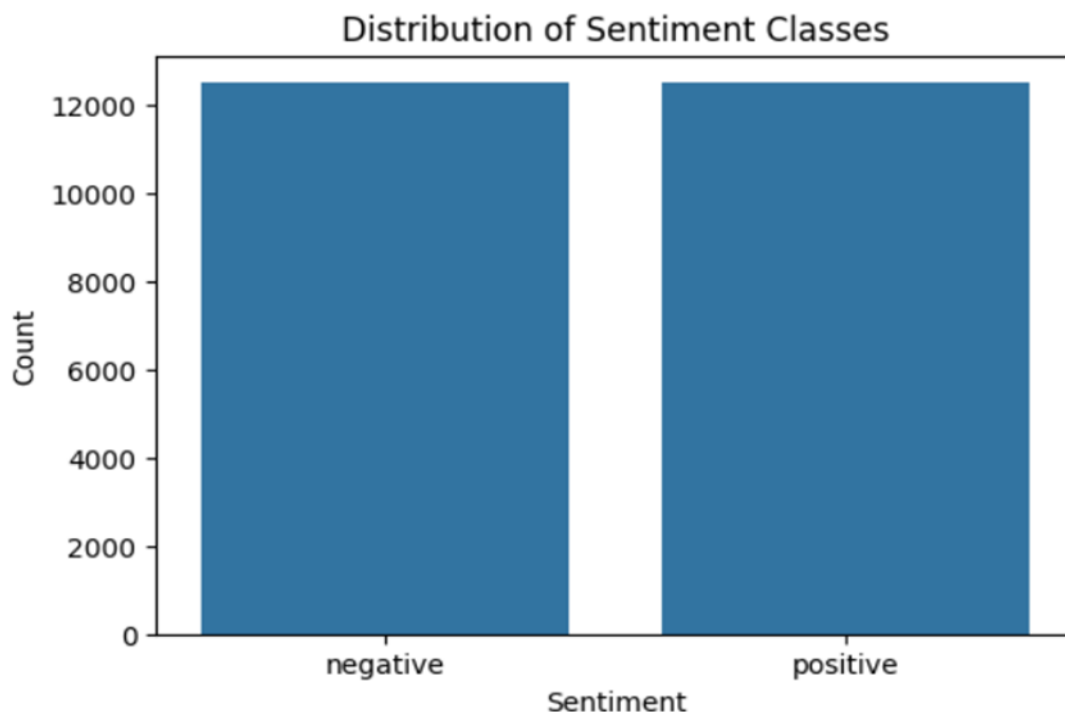


Figure 1: Dataset class distribution



Word clouds reveal common words in positive and negative reviews, visually demonstrating the qualitative aspect of the review texts. However, this information was not used for creating the models but rather aided through the initial data examination.

Before training, the dataset distribution was visualized to check the class balance and to make sure the evaluation metrics would not be affected by class imbalance. Both the train and test sets had the same number of positive and negative reviews.

4. Methodology and Implementation

In order to make the experiments more reproducible and ensure that the variations between runs were minimized, fixed random seeds were utilized whenever possible. The random seeds for Python, NumPy, TensorFlow, and PyTorch were set to yield results that were not only more stable but also more consistent throughout the different executions of the notebooks.

4.1 Text Preprocessing

Initially, the preprocessing pipeline turned all reviews into lowercase, cleaned HTML tags and special characters, broke down the text into words, eliminated stop words, and finally did lemmatization. Negation words like no, not and never have been purposely kept since sentiment classification greatly depends on them. For instance, phrases like "not good" would literally get their meaning changed if the negations are removed.

The text after cleaning was used to build models based on TF-IDF. DistilBERT was run on the original review text, since transformer models are meant to understand the context of language directly and therefore do not require the kind of manual preprocessing that classical models do.

4.2 Feature Encoding

Text data for classical machine learning algorithms and the deep learning model was encoded through TF-IDF up to a maximum of 5, 000 features. Due to the fact that TF-IDF penalizes the terms occurring all over the documents and at the same time highlights those which are quite informative in a particular document, it is a very good choice here. Together with linear models, TF-IDF representations, that come as high-dimensional and mostly zeroes, deliver especially great performance.

4.3 Models

In total, four groups of models were tried. Naive Bayes was chosen as a basic probabilistic model just to have something simple and easy to beat. Linear SVM was the model of choice for text classification of high-dimensional features as it is one of the most used methods out there. In order to reduce overfitting and improve generalization performance, we implemented a feed-forward neural network using TF-IDF features along with dropout regularization and early stopping. Lastly, the DistilBERT model was used as a transformer based one which can figure out the context between words rather than focusing on individual word occurrences only. So it is capable of understanding semantics way better than usual TF-IDF methods.

Within this work DistilBERT is a pre-trained and SST-2 fine-tuned model. So, instead of it being a model fine-tuned specifically on IMDB reviews, it is considered a strong advanced baseline. It is important to take this into account when analyzing the results.

To enhance reproducibility, fixed random seeds for NumPy, Tensorflow, and Python random operations were set where appropriate.

5. Model Evaluation and Results

The evaluation metrics include accuracy, precision, recall, and F1-score. Accuracy is a metric that checks the overall correctness of the model. Precision indicates the percentage of predicted positive reviews that are truly positive. Recall reveals the proportion of actual positive reviews that are identified. F1-score is a harmonic mean of precision and recall and reflects both types of errors, false positives and false negatives, hence it is the primary metric for comparison.

Table 3: Progressive evaluation of Naive Bayes and SVM

Model / Scenario	Train size	Accuracy	Precision	Recall	F1-score
Naive Bayes (Small)	1,000	0.7872	0.8706	0.6746	0.7602
SVM (Small)	1,000	0.8120	0.8052	0.8230	0.8140
Naive Bayes (Medium)	10,000	0.8381	0.8580	0.8104	0.8335
SVM (Medium)	10,000	0.8514	0.8604	0.8389	0.8495
Naive Bayes (Full)	25,000	0.8410	0.8552	0.8210	0.8377
SVM (Full)	25,000	0.8619	0.8676	0.8541	0.8608

The gradual assessment reveals the fact that the traditional both of the models still have a room for improvement when it comes to the amount of training data they use. SVM wins over Naive Bayes in each of the three cases. Well, it lies on the surface because in general, linear SVMs tend to deliver very good performance on sparse TF-IDF features whereas with Naive Bayes, one of the big disadvantages comes from its independence assumption.

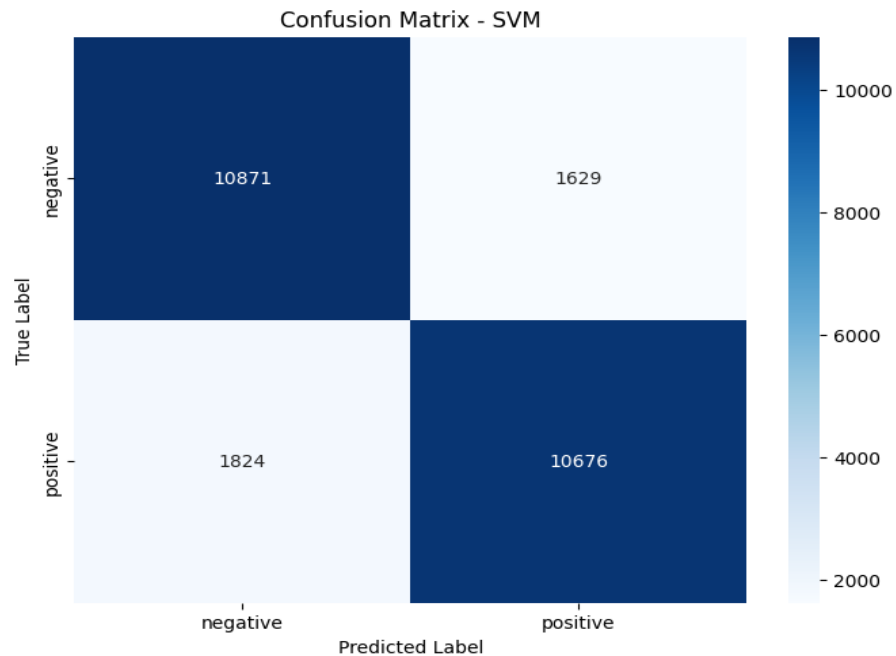


Figure 3: Confusion matrix for the SVM classifier

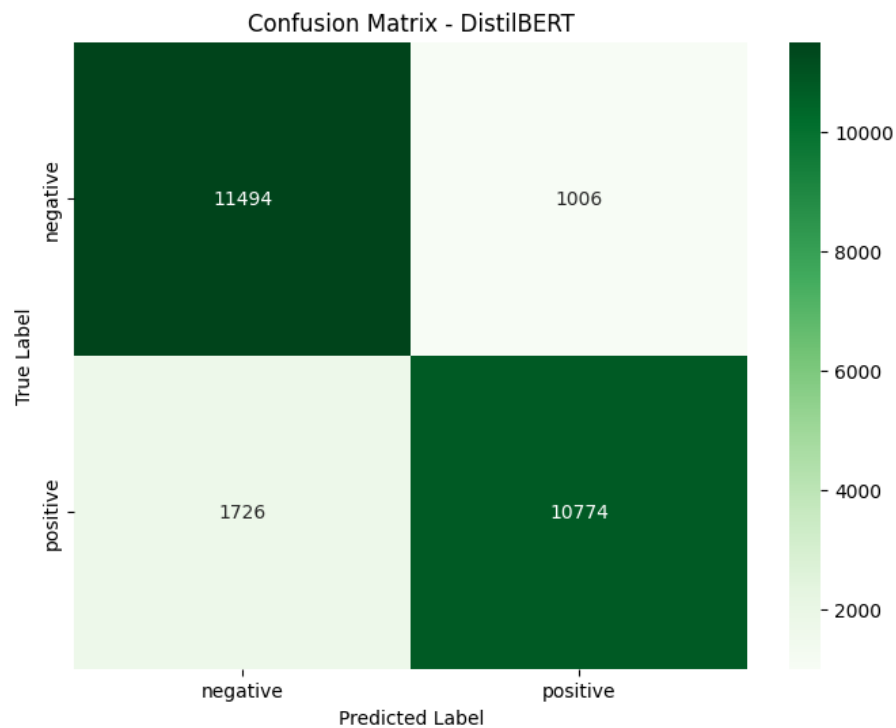


Figure 4: Confusion matrix for the DistilBERT classifier

The confusion matrices illustrate how correct and wrong predictions are spread by the strongest traditional model and the strongest overall model. DistilBERT, on the other hand, has a stronger positive-class precision and a top F1-score.

Table 4: Final comparison of all evaluated models on the full IMDB test set

Model	Accuracy	Precision	Recall	F1-score
DistilBERT (Full Test Set)	0.8907	0.9146	0.8619	0.8875
SVM (Full Test Set)	0.8619	0.8676	0.8541	0.8608
Neural Network (Full Test Set)	0.8663	0.8974	0.8271	0.8608
Naive Bayes (Full Test Set)	0.8410	0.8552	0.8210	0.8377

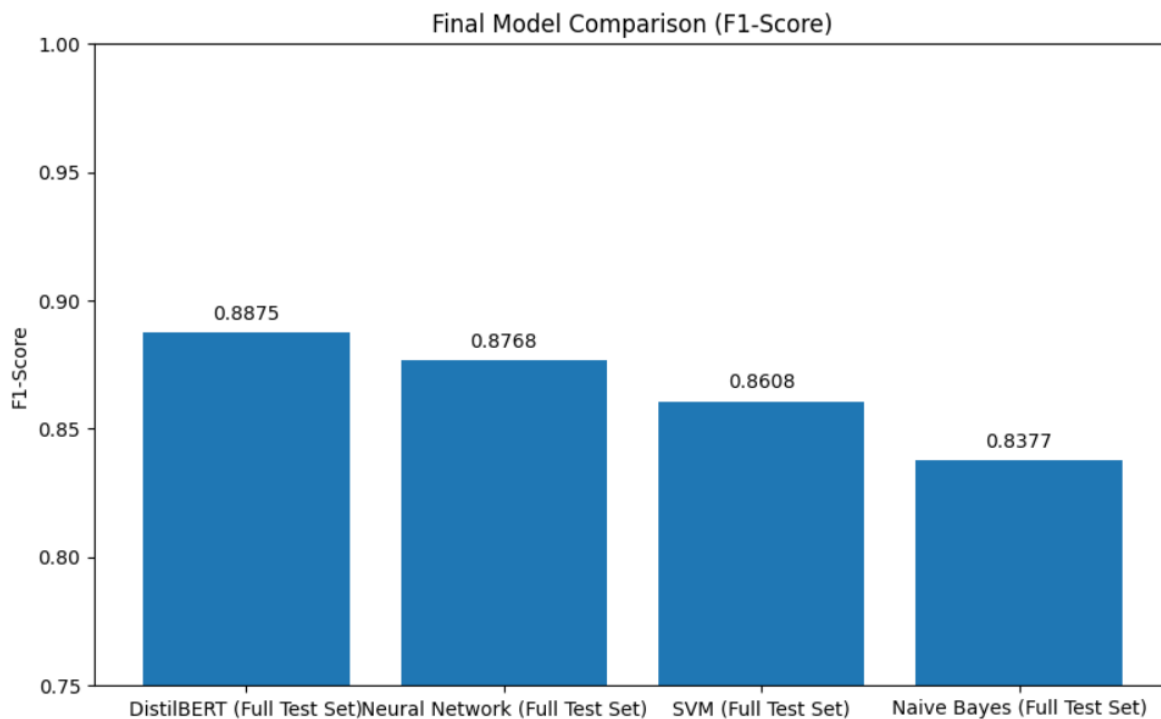
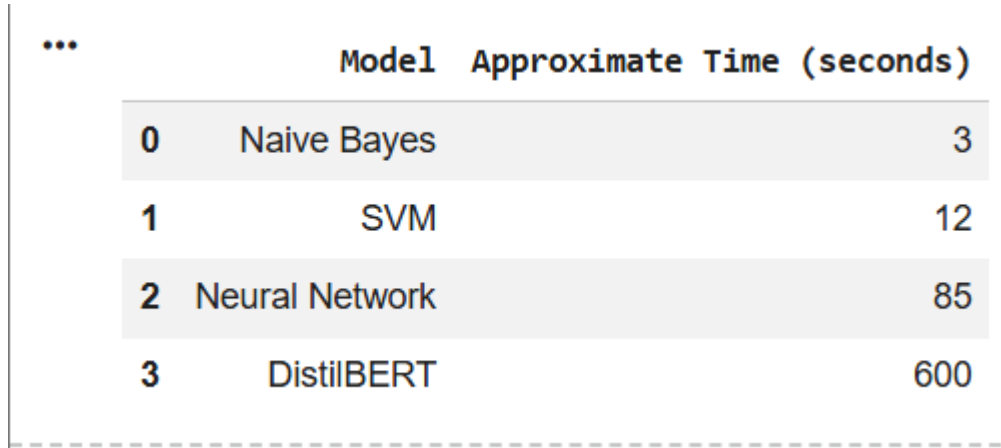


Figure 5: Visualization of the final model comparison

The best overall results are reached by DistilBERT with an F1score of 0.8875 and accuracy of 0.8907. This result emphasizes that contextual representations with transformers are definitely more potent for this sentiment-analysis work as compared to TF-IDF driven models. The neural network and SVM obtain the same F1-score of 0.8608, but their precision-recall trade-offs differ. The neural network manages to get higher precision whereas recall is lower than SVM's, on the other hand, SVM gives a more balanced traditional baseline. Naive Bayes is still quite competitive but at the same time, it is the weakest final model.

Besides predictive performance, execution time was also taken into consideration. Traditional TF-IDF models like Naive Bayes and SVM took much less time to train and predict than DistilBERT. Nevertheless, DistilBERT not only got the highest overall accuracy and F1-score but also showed the usual trade-off between computation cost and predictive performance in NLP systems.



...		
	Model	Approximate Time (seconds)
0	Naive Bayes	3
1	SVM	12
2	Neural Network	85
3	DistilBERT	600

Figure 6: Performance/Execution time of the models visualization

The neural network's learning process was marked by validation instability. This is a known phenomenon when dense neural networks are trained on sparse TF-IDF features. Dropout and early stopping were implemented as a means to deal with overfitting. However, the experiment result serves as a confirmation that the transformer-based DistilBERT model is better fitted for this sentiment-analysis task.

From the applicability viewpoint, if low computational cost is a key factor, the SVM model strategy is really worth considering, because it not only produces competitive results but also has a short training time. On the other hand, DistilBERT will be a better choice if the main focus is on the highest accuracy, although more computational resources will be needed. Such a work-performance tradeoff is quite essential for durable deployment: a light-weight model would fulfill the aim of a swiftly operated first stage screening while a transformer model stands for higher-stakes cases leading to in-depth sentiment monitoring.

The end result therefore is a combination of a single model and an elaborated documented comparison of alternative model choices. Moreover, the comparison presents detailed information in support of the choice of model based on the resource availability and performance requirements.

6. Application Scenario

For the sake of showing the practical utility, the results of the training were exposed to several fresh movie reviews depicting the scenarios of positive, negative, and mixed sentiments. Moreover, DistilBERT gave the confidence scores of every prediction, which is pretty handy in the cases of the real world where a certain prediction is essential.

One of the scenarios of deployment could be the review aggregation platforms, streaming services, and social-media monitoring systems where it is a challenge to analyzing large volumes of the user opinions in an automatic and efficient way.

As a result, the system that has been created shows the capability of changing according to the different requirements of the user. Means of using the Traditional models like SVM if only a light and computationally efficient sentiment screening is to be done whereas the transformer-based approaches like DistilBERT are for the ones that in the need high performing and contextually understanding predictive applications.

This exemplifies that the developed system is flexible in adaptation according to deployment constraints: Use of SVM for lightweight and fast sentiment screening and DistilBERT for more deeply analysed sentiment monitoring.

Table 5: Example application outputs for new movie reviews

Example review	Predicted sentiment	Confidence
This movie was absolutely fantastic! Best film of the year.	Positive	99.8%
Terrible acting and boring plot. Waste of time.	Negative	99.7%
The cinematography was stunning but the story was weak.	Negative	82.4%
I laughed and cried - a true masterpiece!	Positive	99.9%
Surprisingly emotional and visually beautiful.	positive	97.6%

The example predictions demonstrate the intended product of the project: a binary classification system that receives a movie review and outputs positive or negative sentiment.

7. Reflection and Limitations

One of the foremost merits of the project is the very fair final comparative analysis. The high-end transformer model was tested on the comprehensive 25, 000-review test set, instead of a tiny sample. This substantially increases the reliability of the final comparison and eliminates the possibility of performance exaggeration.

Theoretically, one of the challenges of implementing the idea would have been the long inference time of DistilBERT when running on CPU-only hardware. This problem was resolved through Google Colab GPU acceleration, which allowed the full-test-set evaluation to be done within the project period. Besides, this move could be seen as making the comparison more equitable given that the advanced model was able to be run against the same full test set as the rest of the models.

One of the limitations of the study is the use of DistilBERT as the pre-trained transformer baseline only, which was not fine-tuned specifically for the IMDB dataset because of the constraints in computational power and time. It basically served as a pre-trained sentiment model with SST-2 as the fine-tuning dataset. Directly fine-tuning DistilBERT on the IMDB training data might lead to better results, yet it would mean more time spent on training and more computational resources. Another disadvantage is that the simple neural network, which uses TF-IDF features, is not capable of representing word order and contextual meaning as well as a transformer model.

Working on Google Colab with GPU acceleration allowed us to significantly speed up the DistilBERT inference time. Hence, conducting evaluation on the entire 25, 000-test set of reviews became technically doable in the scope of this project.

Further analyzing the errors, the authors carried out an error-analysis step to check examples of the misclassified reviews. A lot of incorrect predictions were related to sarcasm, mixed sentiment, or context-dependent language. It brings out the point that advanced transformer-based NLP systems are still at a loss when the highly nuanced human language is concerned.

How the inputs are treated also determines the output. Retaining negation words helps maintain the original sentiment, however, overzealous discarding of words can still lead to a loss of subtle contextual information. Hence, DistilBERT, which operates on original texts, is able to outperform models that rely on manually cleaned TF-IDF features.

8. Conclusion

This project illustrates an entire NLP sentiment-analysis system designed for movie reviews. The output reveals that DistilBERT best fits this job, earning the top accuracy and F1-score for the entire IMDB test set. This backs up the finding that transformer-based contextual embeddings work better for sentiment classification than TF-IDF based approaches.

Of the classical machine learning models, SVM wins the race and sets a strong baseline as it is both efficient and capable on sparse high-dimensional text features. Naive Bayes is a neat and good baseline, however, the independence assumption hampers its performance. Although the neural network shows good results, it still is less than DistilBERT, proving that both the model architecture and representation quality matter in NLP.

As for future work, the primary enhancement will be fine-tuning a transformer model on the IMDB training set and contrasting it with the pre-trained DistilBERT pipeline. More trials of hyperparameter tuning, cross-validation, and error analysis could make the system even more reliable.

The project, in general, shows the ability to translate theoretical knowledge into practice with the help of real-world data somehow, but it also applies a reasoned criticism of the existing NLP-based methods for analyzing sentiment.

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Appendix

Appendix A - Repository link: [BrailleNshuti/Project-NLP-DLBAIPNLP01-Sentiment-Analysis-of-Movie-Reviews](https://github.com/BrailleNshuti/Project-NLP-DLBAIPNLP01-Sentiment-Analysis-of-Movie-Reviews)

Appendix B - Notebook: PROJECT_NLP.ipynb contains the full executable implementation and outputs used in this report.

Appendix C - Additional screenshots and visualizations from the notebook execution are presented in the submitted PDF document complementary to Appendix C.